

**CODERAXO – WEB DEVELOPMENT INTERNSHIP
PROGRAM**

Final Internship Report

Project & Team Collaboration System

Intern: Aqsa Aslam

Program: CoderAxo Web Development Internship

Offer ID: CAX-OL-2026-310

Internship Duration: June 15, 2026 – August 15, 2026

Acknowledgement

I would like to express my sincere gratitude to the CoderAxo Web Development Internship team for the opportunity to complete this eight-week internship and for the structured, hands-on mentorship provided throughout the program. I am especially thankful to the mentors who led the live Zoom sessions each week and were consistently available to clarify doubts, review my submissions, and keep the cohort on track.

I would also like to thank my family and friends for their patience and encouragement during the busier weeks of the program, particularly around the Final Project deadline, and my institution for supporting and recognising this internship. This experience has strengthened both my technical foundation and my professional habits, and I am grateful for every part of it.

Executive Summary

This report summarises my eight-week Web Development Internship with CoderAxo (Offer ID: CAX-OL-2026-310), completed between June 15 and August 15, 2026. The internship was delivered through live Zoom sessions covering HTML, CSS, JavaScript, Node.js, Express.js, React, and MongoDB, each followed by independent practice and hands-on tasks of increasing difficulty.

Over the eight weeks I completed three progressively harder individual tasks — a personal portfolio website, a To-Do application with local data persistence, and a deployed React countries-explorer app — before applying everything learned to a ten-day Final Project: a full-stack MERN application called the Project & Team Collaboration System. It includes authentication, a dashboard, team and project management, a Kanban-style task board, file sharing, and notifications, all built and secured end to end.

Alongside the technical work, the internship placed strong emphasis on documentation, weekly activity logs, progress reports, and meeting deadlines, closely mirroring how work is managed on a real development team. By the end of the program I had built a practical, working foundation across the MERN stack, strengthened my time-management and documentation skills, and produced a complete project I can reference in interviews and on my portfolio going forward.

Table of Contents

Acknowledgement	2
Executive Summary	3
1. Introduction	5
2. Internship Details	5
3. Work Samples / Project Summaries	7
4. Learning Experiences	8
5. Challenges & Solutions	9
6. Reflection & Conclusion	10
Appendices	11

1. Introduction

This report covers the eight weeks I spent as a Web Development Intern at CoderAxo, from June 15 to August 15, 2026. I joined the program to build a proper, structured foundation in web development — something beyond what a single university course usually covers — and to gain hands-on experience with the MERN stack (MongoDB, Express.js, React, and Node.js) used by most real-world web teams today.

CoderAxo is a cohort-based training and internship program built around practical, project-driven learning. The Web Development track I joined is structured around the core technologies used to build modern full-stack applications: HTML, CSS, and JavaScript on the front end; Node.js and Express on the back end; React for interactive interfaces; and MongoDB as the database layer. Alongside the technical sessions, the program places significant weight on documentation, deadlines, and submission discipline — every task required a clean repository, a written report, and, for the Final Project, a recorded demonstration — which closely mirrors how work is reviewed in a professional setting.

Going into the internship, my goals were to build a genuinely practical (not just theoretical) foundation in HTML, CSS, and JavaScript; get comfortable with backend development using Node.js and Express; learn React well enough to build real, multi-page, API-connected applications; work through a complete MERN-stack project the way it would be handled on an actual team; and improve how I document, structure, and present my own work. By the end of the eight weeks, I felt all five goals were genuinely met, with the Final Project serving as proof — it required every one of those skills at once, under a real deadline.

2. Internship Details

2.1 Duration & Timeline

The internship ran for eight weeks, from June 15 to August 15, 2026, structured as a clear progression from front-end fundamentals to a complete full-stack build. Each week combined a live Zoom session with independent practice, a written activity log, and, for several weeks, a graded hands-on task.

Week	Dates	Focus
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1	Jun 15 – Jun 21	Onboarding; HTML5 fundamentals (semantic structure, forms, tables)
2	Jun 22 – Jun 28	CSS fundamentals (box model, Flexbox, Grid, positioning, responsive design)
3	Jun 29 – Jul 5	JavaScript fundamentals; Task 1 assigned (Personal Portfolio)
4	Jul 6 – Jul 12	Task 2 – To-Do App (advanced JavaScript, localStorage persistence)
5	Jul 13 – Jul 19	Introduction to Node.js, npm, and core modules; first HTTP server
6	Jul 20 – Jul 26	React fundamentals; Final Project brief released (Jul 25)
7	Jul 27 – Aug 2	Task 3 – React Countries App; Final Project build begins (Authentication)
8	Aug 3 – Aug 9	Final Project completion – Dashboard, Teams, Projects, Task Board, Files, Notifications

2.2 Nature of Work

The work was a mix of guided learning and independent building. Each live session introduced a specific topic — a language feature, a framework concept, or a backend pattern — which I then had to apply directly, either through short practice exercises or a full graded task. This progressed from static front-end pages in Week 1 to a fully authenticated, database-backed MERN application by Week 8. Every task required a working, deployed or runnable deliverable, not just code that compiled, along with a short write-up describing what was built and how.

2.3 Team Interaction

Sessions were delivered live by CoderAxo mentors to a cohort of interns, with direct Q&A rather than pre-recorded content. Mentors reviewed weekly activity logs and progress reports and were reachable for clarification between sessions. While the individual tasks (portfolio, To-Do app, React app) were completed independently, the Final Project brief was written and reviewed the way a real product requirement would be — with a fixed scope, a hard deadline, and defined submission standards (source code, README, ER diagram, API documentation, and a demo video) that had to be met without ambiguity.

2.4 Tools & Technologies

- Front end: HTML5, CSS3 (Flexbox, Grid), JavaScript (ES6), React 18, React Router, Tailwind CSS, Vite
- Back end: Node.js, Express.js, REST API design
- Database: MongoDB with the Mongoose ODM, MongoDB Atlas
- Authentication & Security: JWT-based authentication, bcrypt.js password hashing, protected routes, input validation
- Supporting libraries: Axios, Framer Motion, Recharts, Lucide Icons
- Tooling: Git and GitHub for version control, npm for dependency management, VS Code as the primary editor, Zoom for live sessions

3. Work Samples / Project Summaries

3.1 Task 2 – To-Do Application

Assigned in Week 4, this task asked for a fully functional To-Do application built with vanilla JavaScript. I implemented task creation and deletion, a 'mark as completed' toggle, a dark/light mode switch, and data persistence using localStorage, so the task list survived a page refresh. The main challenge was keeping the on-screen list, the underlying array of tasks, and localStorage all in sync every time something changed; re-rendering the full list on every update turned out to be the simplest and most reliable approach, and it is a pattern I have reused since.

3.2 Task 3 – React Countries Explorer

Assigned in Week 7, this task required a complete, deployed React application with functional routing, real API data, and proper styling. I built a countries-explorer app with a Home page, individual country detail pages, an About page, a Contact page, and a 404/error page, all connected through React Router, fetching live country data, and styled responsively with Tailwind CSS. This was the first project in the program that combined React state, routing, and a live external API in one working, publicly deployed application.

3.3 Final Project – Project & Team Collaboration System

The Final Project was the centrepiece of the internship: a full-stack MERN application built to help teams plan, assign, and track work in a single dashboard, built and submitted within a ten-day window after the brief was released on July 25. Its core modules were: Authentication &

Account Management (JWT and bcrypt-hashed passwords), a Dashboard with summary cards, task-status distribution, and a recent-activity feed, Team Management, full CRUD Project Management with status tracking, a Kanban-style Task Board (Todo, In Progress, Review, Completed) with priorities and due dates, a File Sharing module for project documents, and a Notifications module for team and task activity.

I built the project using a vertical-slice approach — completing the backend and frontend for one module together before moving to the next, starting with Authentication — which meant there was always a working application at every stage of the ten days rather than a pile of disconnected pieces coming together only at the end. On the database side, MongoDB collections were normalised and linked through ObjectId references: a team can have multiple members and projects, each project holds multiple tasks assigned to individual users, and activities and notifications reference back to the relevant users, projects, or tasks. For submission I prepared the complete source code, a database export, a GitHub repository, a README with installation instructions, screenshots, an ER diagram, API documentation, and a demo video, and submitted the project ahead of the August 5 deadline.

4. Learning Experiences

4.1 Knowledge Gained

I moved from writing semantic HTML and basic CSS layouts to understanding how a complete full-stack application fits together — how a React front end talks to an Express API, how that API validates requests and talks to MongoDB, and how authentication tokens tie the two sides together securely. Building the raw Node.js HTTP server before Express also gave me a clearer picture of what a framework is actually doing underneath its routing and middleware.

4.2 Skills Gained

- Semantic, accessible HTML5 and responsive CSS using Flexbox and Grid together
- Core JavaScript — DOM manipulation, event handling, arrays/objects, and ES6 syntax
- Server-side fundamentals with Node.js and REST API design using Express.js
- Building component-based, stateful interfaces in React, including hooks and React Router

- Full MERN-stack integration, connecting a React front end to an Express/MongoDB back end end to end
- Authentication and security fundamentals — JWT, bcrypt password hashing, protected routes, input validation
- Using Git and GitHub for version control and deploying front-end applications publicly

4.3 Values

Beyond the technical stack, the internship reinforced habits that matter on any real team: meeting weekly deadlines consistently, documenting work clearly enough that someone who has never seen the code could pick it up, and being honest about what could and could not be finished well in a fixed window rather than quietly cutting corners. Working through a hard ten-day deadline also reinforced the value of planning before building rather than starting to code immediately.

4.4 Most Challenging Task

The most challenging part of the internship was pacing the Final Project. Ten days is not a lot of time for a full MERN application with eight modules, authentication, and security requirements, especially while Task 3 was still due in the same stretch. The vertical-slice approach — finishing Authentication completely before moving on, rather than half-building every module at once — was what made the deadline realistic, since it guaranteed a working, demonstrable product at every stage rather than a set of unfinished pieces.

5. Challenges & Solutions

A few specific challenges stood out across the eight weeks. Early on, in Week 2, a Flexbox navbar that would not centre vertically turned out to be caused by a missing height on its parent container — a small mistake that took longer than it should have to spot, and one that made me more careful about checking parent context before debugging child styles.

In Week 4, keeping the To-Do App's task list synced correctly across localStorage, the underlying data array, and the UI took a few attempts before I settled on simply re-rendering the whole list on every change, which was simpler and more reliable than trying to patch individual DOM nodes.

The bigger challenge, as noted above, was pacing during the Final Project, where a new topic, a task, and the Final Project brief all landed close together in Weeks 6 and 7. The solution was the same vertical-slice discipline: build and stabilise one module completely, in priority order, before starting the next, so that time pressure reduced the polish on later modules rather than leaving the whole application unfinished.

6. Reflection & Conclusion

Looking back at these eight weeks, what stands out most is how much changed between the version of me writing my first HTML tags in Week 1 and the version who shipped a full MERN-stack Team Collaboration System in Week 8. I have done web development coursework before, but this was the first time I built through a topic every single week and then immediately had to use it in something real — a portfolio, a To-Do app, a deployed React project, and finally a complete application under a hard deadline. That rhythm is what made the concepts stick.

If I am honest about what I would do differently, it is time management around documentation — I kept writing READMEs and reports until the last day of each task instead of documenting as I went, and would manage that more deliberately next time. The skills I am taking forward are less about any single technology and more about the workflow itself: plan the schema before writing code, build in working slices instead of half-finishes everything at once, and document as I go rather than after.

My longer-term goal is to move toward a Java Developer role while keeping my web development skills sharp through freelancing, and this internship strengthened exactly the JavaScript, React, and backend fundamentals that carry over into that goal, while giving me a complete project to reference in interviews and on my portfolio. I am grateful to the CoderAxe team for the structure, the mentorship, and the push to build something real rather than just learn in theory.

Appendices

Appendix A: Weekly Activity Logs

Weekly Activity Logs (Weeks 1–8), detailing the sessions attended and work completed each week, are included as separate supporting documents alongside this report.

Appendix B: Task & Project Screenshots

Screenshots of Task 3 (React Countries Explorer) and the Final Project (Project & Team Collaboration System) are included as separate supporting files alongside this report.

Appendix C: Final Project Supporting Documentation

The Final Project submission package includes the complete source code, a database export, a GitHub repository link, a README with installation instructions, an ER diagram, API documentation, and a recorded demo video, referenced via a Google Drive link in the accompanying Final Project Report.

Appendix D: Offer Letter & Completion Letter

Copies of the internship offer letter (CAX-OL-2026-310) and the completion letter issued by CoderAxo are included as separate supporting documents alongside this report.